UNIT-II

Process

- What to call all CPU activities?

- Batch system – jobs
- Time-shared system – tasks

- All these activities are called as processes.

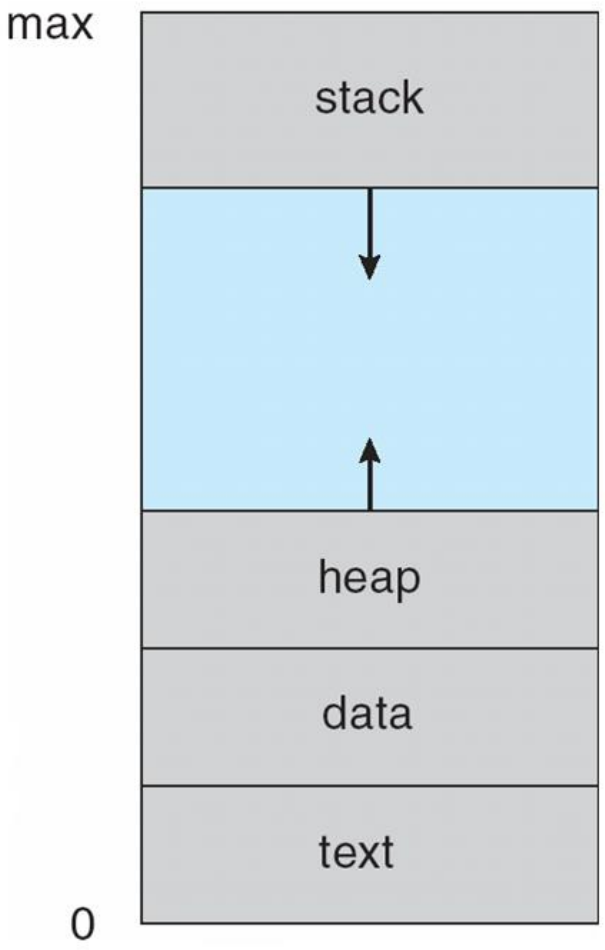
The Process

- A process is a program in execution.

- A process includes text section, data section, heap and stack.

- Text section – program code
- Data section – global variables
- Heap – dynamically allocated memory
- Stack – contains temporary data such as function parameters, return address and local variables.

Process in Memory



- A program is not a process.
- A program is a passive entity.
- A process is an active entity.
- A program becomes a process when an executable file is loaded into memory.
- By double-clicking icon or entering name of file.

- A process itself can be an execution environment for other code.

- Example: java programming environment
- An executable java program is executed within the java virtual machine.

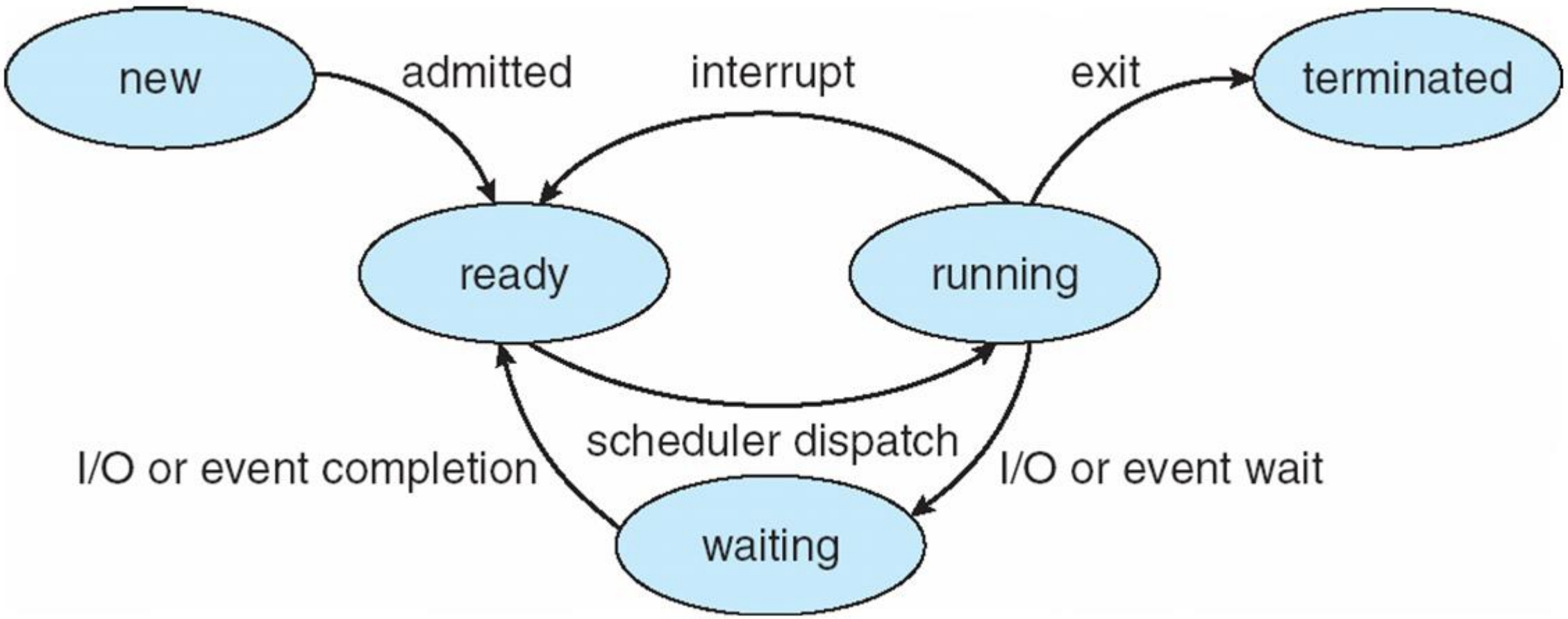
Process State

- As a process executes, it changes state.

- State – current activity of the process.

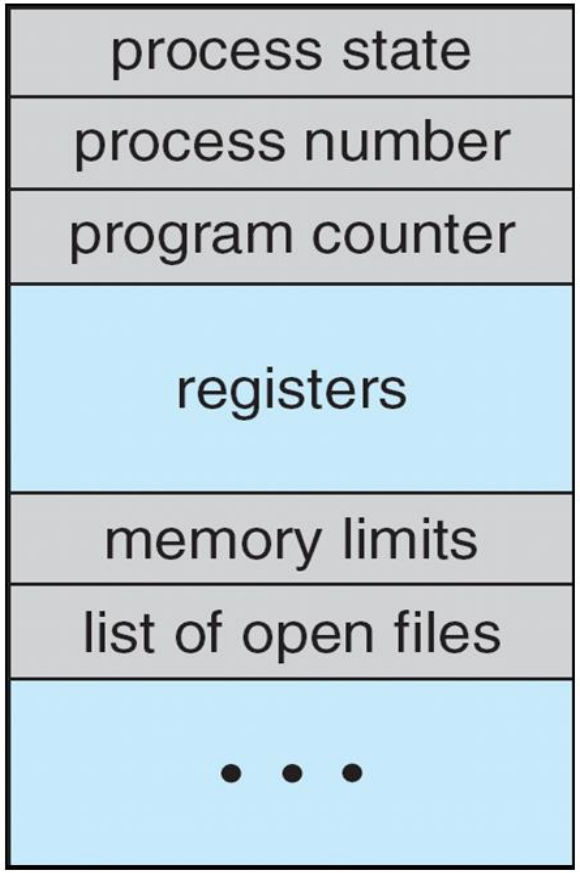
- New – created.
- Running – executed.
- Waiting – waiting for some event.
- Ready – waiting to be assigned to a processor.
- Terminated – finished execution.

Process State



Process Control Block

- Each process is represented in the OS by PCB.



PCB

- Process State
- Program Counter
- CPU registers – accumulators, index registers, stack pointers, general-purpose registers.
- CPU-Scheduling information – process priority, pointes to scheduling queues and scheduling parameters.
- Memory-mgt infn – value of the base and limit registers, page tables, segment tables.
- Accounting infn – amount of CPU used, time limits, account numbers, process numbers
- I/O Status – list of I/O devices allocated to a process,open files.

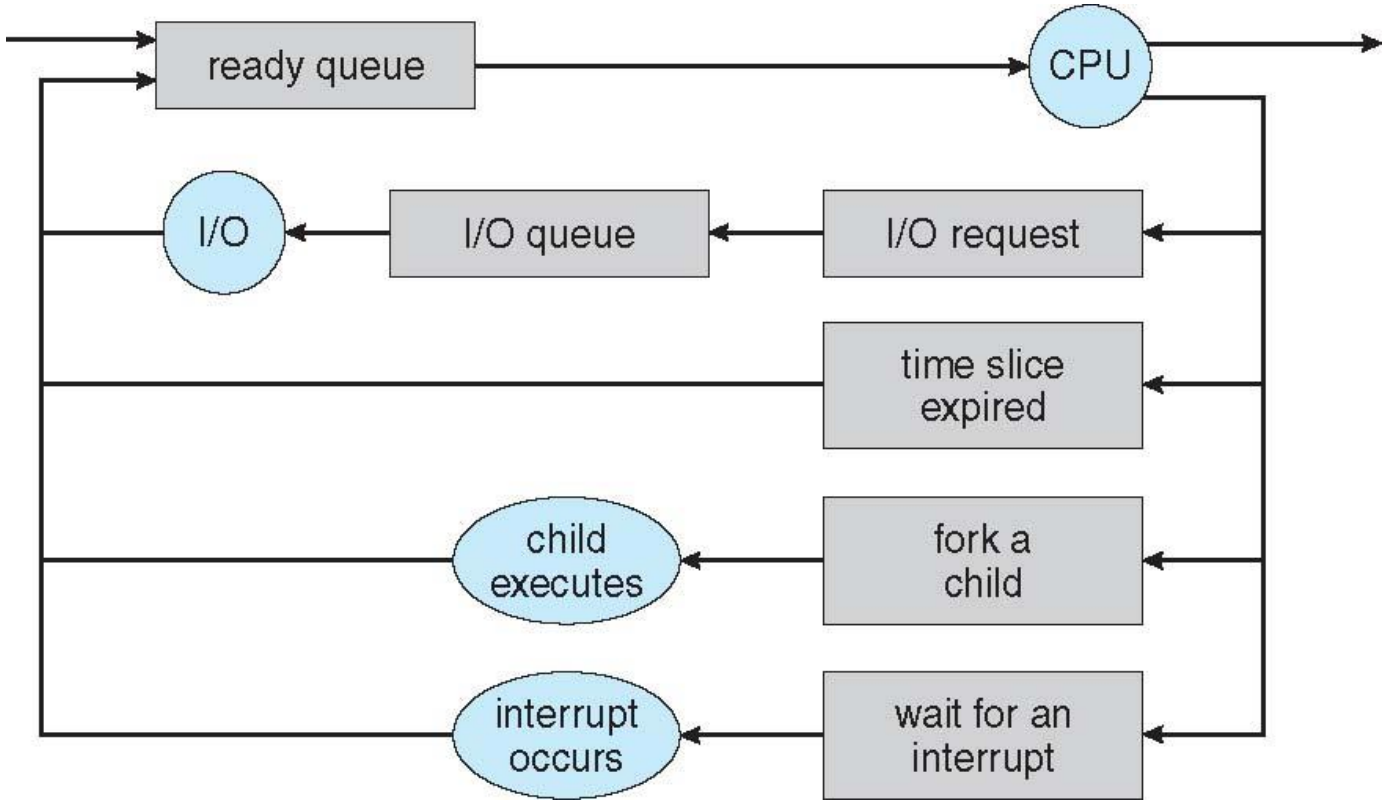
Threads

- A process is a program that performs a single thread of execution.
- Example: Word processor – type characters and run spell check.
- Modern operating systems – multiple threads of execution – more task at one time.

Process Scheduling

- Process scheduler – selects an available process for program execution on the CPU.
- When the process enter the system, they are put into a job queue.
- The process that are ready and waiting to execute are kept in the ready queue.
- List of processes waiting for a particular I/O device is put on device queue.

Queuing Diagram



Schedulers

- A process migrates among the various scheduling queues throughout its lifetime.

- Scheduler – responsible for selecting the process for scheduling.
- Long-term scheduler
- Short-term scheduler
- Medium-term scheduler

- Batch system – processes are spooled to mass storage device.
- Long-term scheduler – selects processes from this pool and load into memory.
- Short-term scheduler – selects from among the processes that are ready to execute and allocates the CPU to one of them.

- Short-term scheduler – executes frequently.
- It must be fast.

- Long-term scheduler – executes much less frequently.
- Is invoked only when the process leaves the system.
- Can take long time to decide which process should be selected for execution.
- Careful selection.

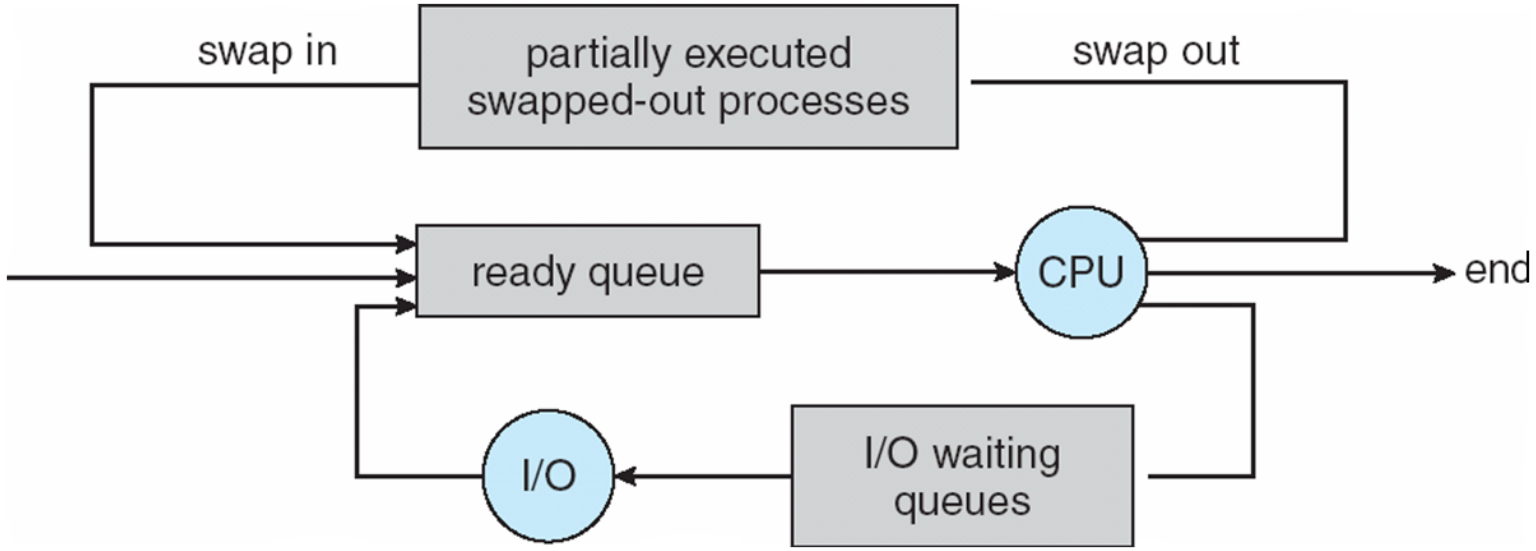
- Process – I/O bound or CPU bound.

- I/O bound – spend more time on I/O
- CPU bound – spends more time on computations.

- LTS – select a good process mix of I/O bound and CPU-bound – good system performance.

- Some operating systems have additional intermediate level of scheduling – medium-term scheduler.

- Medium-term scheduler – swapping – swap-in and swap-out process – to free up memory – improve the process mix.



Context Switch

- When an interrupt occurs, the system needs to save the context of the process running on the CPU.
- The context is represented in the PCB of the process.
- It includes the value of CPU registers, the process state, and memory-management information.
- Switching the CPU to another process requires performing a state save of the current process and a state restore of the different process. This is known as context switch.

- Context switch – saves the context of old process - Loads the context of the new process scheduled to run.

- Context-switch time is pure overhead.

- Context-switch time are highly dependent on hardware support.